Ebna Sina



Data Scientist · MSc Data Science student, Berlin

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Data science master's student with 4+ years across software engineering, ML research and research data infrastructure. Co-author of an IEEE Access paper on real-time vehicle detection. Looking for a **working-student role or master's thesis in ML / LLMs**: 20 hours a week on a student visa, or a switch to a work visa for a full-time role.

EXPERIENCE

Dec 2025 – present
Working Student, Research Data Management · Paul-Drude-Institut (Leibniz), Berlin
- Operate the institute's self-hosted **NOMAD Oasis** research data platform: Docker, Nginx/HTTPS, Grafana/Prometheus monitoring, CI.
- Extend its Python plugin with schemas, normalizers and search apps for MBE growth and sample data; contribute upstream to FAIRmat.

Jun 2024 – Apr 2025
Research Assistant · United International University, Dhaka
- Benchmarked **11 pre-trained CNNs** on 6 datasets and built a **54,556-object** dataset of Bangladeshi road vehicles.
- Deployed YOLOv8 and SSD-MobileNetV2 on an NVIDIA Jetson Nano: **93% detection rate, 98% accuracy** in live Dhaka traffic. Published in IEEE Access.

Sep 2021 – Nov 2024
Software Engineer · Orange Solutions Ltd (United Group), Dhaka
- Built data-visualisation dashboards and analysed business data across full-cycle ERP modules: production, sales, inventory, finance, SCM, HR & payroll.
- Promoted from Junior Software Engineer in January 2024.

SELECTED PROJECTS

2026
FootScout · player-similarity model on real match events · ebnasina7.github.io
- Built from StatsBomb event data: **714 players, 250 matches**, 19 per-90 rates plus touch heatmaps. In a split-half test, a player's other half ranks in the top 5 for **29% of 553 players, against 1% by chance**.

2026
Hybrid Brain · LLM routing
- A small decision model routes each task to specialist LLMs on the GWDG supercomputer and escalates hard cases to a 675B model; **20–30× faster** than LLM-as-router, runs locally on llama.cpp.

PUBLICATION

D. M. Farid, P. K. Das, M. Islam, **E. Sina**. "Bangladeshi Vehicle Classification and Detection Using Deep Convolutional Neural Networks With Transfer Learning." *IEEE Access*, vol. 13, pp. 26429–26455, 2025. doi:10.1109/ACCESS.2025.3539713

EDUCATION

Oct 2025 – present
MSc Data Science · Berliner Hochschule für Technik (BHT), Berlin
Recommender systems, deep learning, large language models

2016 – 2020
BSc Computer Science & Engineering · United International University, Dhaka
CGPA 3.77/4.00, tuition scholarship · Thesis: Bengali conversational question answering (Bi-LSTM)

SKILLS

Machine learning	Python, PyTorch, scikit-learn, TensorFlow/Keras, Sentence-BERT, Optuna, Weights & Biases
LLMs	llama.cpp, Hugging Face, LoRA, LLM routing & evaluation
Data	pandas, NumPy, PySpark, SQL (PostgreSQL, MySQL, MS SQL), MongoDB, Streamlit, Plotly, Tableau
Infrastructure	Docker, Nginx, Linux, Git, GitHub Actions, Grafana, Prometheus

CERTIFICATES

IBM Data Science Professional Certificate · Dec 2023

Google Data Analytics Professional Certificate · Apr 2023

LANGUAGES & MORE

Bengali (native) · English (fluent) · German (A2)

Football: champions in Darmstadt, runners-up at Desh Elite Cup 2026 (RB Berliners)